

Engineer who loves building, researching, and optimizing high-throughput data systems. Experienced in leading agentic product rollouts through strict evaluation suite design, optimizing model/prompt/inference for real-time work, and debugging complex data infrastructure where performance, reliability, and cost are critical for downstream applications.

Experience

Software Engineer III (ML) **Chartbeat** **Oct 2024–Present**
New York, New York

- Driving the adoption and implementation of agentic & LLM-based products with a focus on evaluation driven development, integrating LLM-tooling into developer workflows, and cost-aware inference and deployment strategies
- Led development and evaluation for a content recommendations service, reduced cost by ~75% by using prompt engineering strategies to replace an expensive RAG system while preserving recommendation quality
- Utilized LLM-based article topic assignment to increase NER pipeline’s accuracy by ~50% with a new eval system
- Designed a <20ms P99 system for feeding Google Ads Manager page data for each tenant’s user’s page load
- Reduced MLOps costs 80% by replacing paid software with comparable, in-house training & monitoring systems
- Cut Snowflake account spend by 10% and decreased lag spikes by 20x through reverse-engineering autoreclustering
- (all roles) Mentored multiple interns through both general engineering day-to-day and impactful projects
- (all roles) Member of on-call rotation, work with Support team to communicate outages & firefight system problems

Software Engineer II (Data) **May 2023–Present**

- Built a conversion event tracker to apply attribution models to hundreds of events per second against a stream of 60k+ page visits/sec, serving them in real-time & historically, surfacing over 250 TB of raw data (Kafka, Postgres, Snowflake)
- Built a holistic health metrics pipeline & dashboard for our primary data pipeline, saving on-call hours of debugging
- Lead investigator for Snowflake & other data system optimization and firefighting problems

Software Engineer I **Oct 2021–Present**

- Doubled the historical product’s features & data range while keeping latency constant via tweaking SF clustering keys
- Optimized many Snowflake and Postgres tasks & tables via reducing join sizes, improving keys, ordering, etc.
- Developed a historical method for gathering clustering depth statistics, using them to locate badly clustered tenants

Software Engineer **Logos** **May 2020–Oct 2021**
WAMI Development team Reston, Virginia

- Developed performant C++ features & bug fixes for a real-time, on-prem, Wide Area Motion Imagery (WAMI) surveillance system with multithreaded processes running distributed across multiple systems
- Handled homographic processing & archival for data streams of ~100 images+metadata/second from multiple sensors
- Communicated directly with operators to determine future features & areas of improvement

Relevant Projects

cleo **Personal ([Github](#), [Hugging Face](#))** **Python, llama.cpp, Qwen3.5**

- Built a unified model & harness to process cheap analytical subagent workloads with a <2B model
- Curated a real high-quality, SQL-based, analytical workflow trace dataset for PeFT, OPD, and RL-based training stages
- Studied tool use patterns to create a minimal-yet-expressive tool kit that can generalize to new workflow paradigms
- Utilized Q4_K_M quantization + self-speculative decoding + multi-token prediction for ~3x perf over base model

Topics Pipeline v2 **Chartbeat** **Python, vLLM**

- Redesigned a topics pipeline that processes >700k news articles/day to cut spend by 33% and boost topic quality

- Created an evaluation suite to measure system throughput, topic quality via a tiered method against a Sonnet 4.6 teacher, and topic stability through time by quantifying mis-attribution to capture the effects on downstream products
- Evaluated over 180 combinations of open source model/quantization, prompt, and vLLM configuration against the eval suite to find Qwen3.5-35B-A3B-FP8 provided 92% of Sonnet 4.6 quality at 1% of the cost through prompt engineering
- Bumped article scrape rates by 10% with an automated bug squashing process that dug through failed scrapes

VQA From Scratch

Personal ([Github](#))

Python, PyTorch

- An attempt to beat frontier Vision-Language models with just my MacBook Pro and (mostly) homegrown datasets
- Applied a formally SSL-based regularization term to VAEs to achieve ~8% better linear probe scores over baseline
- Sourced, cleaned, & tokenized a from-scratch text dataset of over 295M tokens based on the 80 MSCoCo categories
- Found a small, stable LM via probing, scaled cross-attention LR, and finally just using a decoder-only paradigm
- Distilled a subset of wiki articles with Qwen 3.5 into context-question-answer triplets for mixing into later LM batches
- Bridged together the VAE + LM with question-conditioned cross-attention, prefix calibration, and LM adapters

mdlens

Personal ([Github](#))

Rust

- Built a simple, stateless tool to reduce token consumption & tool calls by 66% on Markdown-heavy agentic workloads
- Created an open [evaluation dataset](#) using other Markdown datasets, scraped READMEs, and my own personal docs
- Utilized a heavily agentic, adversarial process by having multiple harnesses & models reach consensus on features
- Works unreasonably well with gpt-5.5 on Hermes Agent to save token cost on my other research projects

SGOCR Dataset

Personal ([Github](#), [Hugging Face](#))

Python, OSS Models

- Designed a spatially-grounded, OCR-focused VQA dataset that taught my small-scale VLM “to read” 12x better
- Utilized an agent-first iteration approach to analyze current pipeline weaknesses, write & test changes, and re-run
- Developed a companion dataset review app for easily grounding the agentic loop on real human feedback
- Hand-crafted evaluation strategies using frontier VLMs for visual feature necessity, question leakiness, and set difficulty
- Tested & picked raw visual-text extraction, anchor discovery & labeling, and teacher methods/models with said evals

Material Segmentation Model

Western Digital (Internship)

Python, Tensorflow

- Designed & tuned a U-Net model that segmented microstructures for the Hard Drive Materials team
- 33% improvement in pixel classification accuracy in 1/20th of the time via local deployment
- Large improvements in accuracy came from batch normalization & inclusion of skip connections (AE -> U-Net)

Education and Publications

B. Sc. Computer Science

Pennsylvania State University
State College, Pennsylvania

2016–2020

- Coursework includes Computer Vision, High Performance Computing, Statistics, Systems Programming, Artificial Intelligence, Database Systems, Entrepreneurship, and Game Theory
- [Learned Neural Iterative Decoding for Lossy Image Compression Systems](#) (Ororbis, et. al., Data Compression Conference (DCC) 2019) Helped publish as a Research Assistant. Built baseline models & located α value.

Technologies and Languages

- Languages: Python, C++, C, Go, SQL, Bash, CUDA
- Technologies: Snowflake, Kafka, PostgreSQL, AWS, Git, Docker & K8s, TF & PyTorch, LLM APIs
- Foreign: French (B1)
- Projects: OSS Database Contributions, ML paper implementations, other side projects are on my [Github](#)